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ORIGINAL RESEARCH

Creatine Supplementation Does Not Influence the Ratio Between Intracellular Water and Skeletal Muscle Mass in Resistance-Trained Men

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The authors aimed to compare the effects of creatine (Cr) supplementation combined with resistance training on skeletal muscle mass (SMM), total body water, intracellular water (ICW), and extracellular water (ECW) in resistance-trained men as well as to determine whether the SMM/ICW ratio changes in response to the use of this ergogenic aid. Twenty-seven resistance-trained men received either Cr ($n = 14$) or placebo ($n = 13$) over 8 weeks. During the same period, subjects performed two split resistance training routines four times per week. SMM was estimated from appendicular lean soft tissue assessed by dual-energy X-ray absorptiometry. Total body water, ICW, and ECW were determined by spectral bioelectrical impedance. Both groups showed improvements ($p < .05$) in SMM, total body water, and ICW, with greater values observed for the Cr group compared with placebo. ECW increased similarly in both groups ($p < .05$). The SMM/ICW ratio did not change in either group ($p > .05$), whereas the SMM/ECW ratio decreased only in the Cr group ($p < .05$). A positive correlation was observed ($p < .05$) between SMM and ICW changes ($r = .71$). The authors' results suggest that the increase in muscle mass induced by Cr combined with resistance training occurs without alteration of the ratio of ICW to SMM in resistance-trained men.

Keywords: DXA, ergogenic aids, muscle hypertrophy, strength training

Muscle hypertrophy is one of the main morphological adaptations resulting from resistance training (RT). Due to the trainability principle, trained individuals exhibit a lesser magnitude of gains in muscle mass than novice or detrained subjects (Ahtiainen et al., 2003; Kraemer & Ratamess, 2004). That is why trained individuals often resort to the use of ergogenic strategies to overcome plateaus and continue to realize meaningful adaptations during an RT program (Helms et al., 2014). Among the plethora of ergogenic options, creatine (Cr) supplementation when combined with RT generally has been shown as an effective aid to enhance muscle mass accretion (Chilibeck et al., 2017; Nissen & Sharp, 2003), particularly in trained individuals (Nunes et al., 2017).

Among the mechanisms by which Cr supplementation can contribute to muscle mass gain, early research proposed that this substance appears to directly affect muscle protein synthesis via modulations of components in the mammalian target of the rapamycin pathway (Farshidfar et al., 2017). Cr supplementation may directly affect the myogenic process by altering secretions of myokines (e.g., myostatin and insulin-like growth factor-1) and the expression of myogenic regulatory factors, resulting in enhanced satellite cells' mitotic activities and differentiation into myonuclei (Farshidfar et al., 2017). Moreover, Cr acts as an osmolyte, inducing an increase in intracellular water (ICW), which, in turn, may increase protein synthesis and, thus, contribute to muscle mass accretion (Farshidfar et al., 2017; Schoenfeld, 2010). The simultaneous analysis of changes in total body water (TBW) and its extracellular water (ECW) and ICW components may help explain the changes that occur in skeletal muscle mass (SMM) in response to Cr supplementation combined with RT.

Results of previous investigations have been inconsistent as to the effects of Cr supplementation combined with RT in relative changes in aspects of body composition. Francaux and Poortmans (1999) observed that Cr supplementation combined with RT induced an increase in absolute values of body mass, TBW, and ICW but not in ECW in untrained men. However, when TBW and ICW were expressed as a percentage of body mass, no significant difference was observed. It should be noted that the experiment mentioned above was conducted with untrained individuals.

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Therefore, considering that the adaptive response to RT is, at least in part, predicated on training status (Ahtiainen et al., 2003), the ability to compare muscular adaptations in untrained individuals may be confounded by the fact that they tend to respond favorably to multiple training stimuli. Moreover, the Francaux and Poortmans (1999) study analyzed relative changes using body mass, which is not an accurate indicator of muscular adaptations.

Powers et al. (2003) revealed that Cr supplementation combined with RT apparently did not induce a significant increase in ICW, but there was an increase in TBW. It is, therefore, conceivable that the increased TBW was due to ECW, although the values were not reported in the study. These results may have been influenced by the inclusion of both men and women in the same sample as well as the response to a different routine in trained individuals. For example, recently, it has been observed that a program that individualizes prescription based on previous training volume elicits significantly greater changes in the cross-sectional area than a standard RT program in resistance-trained men (Scarpelli et al., 2020). Therefore, the internal validity of studies involving individuals with previous RT experience would seemingly benefit from standardizing RT variables among a cohort for a prolonged period before initiating the research protocol.

In this regard, our investigation aimed to compare the effects of RT with Cr supplementation versus placebo (Pla) on SMM, TBW, ICW, and ECW in resistance-trained men who followed a similar preconditioning-RT protocol and the same progression in training volume before the study onset. We also analyzed whether differences occurred in the ICW/SMM ratio from Cr supplementation after 8 weeks of RT. Based on previous studies, we hypothesized that the Cr group would present a higher SMM gain than the Pla group. However, the ratio of ICW to SMM would remain similar within groups.

Methods

Experimental Design

A randomized, double-blinded, and Pla-controlled trial was carried out over 30 weeks. Participants and outcome assessors are all unaware of the intervention assignments during the trial. Testing was carried out at Weeks 1–2, 19–20, and 29–30 and consisted of anthropometric and body composition measurements. A preconditioning RT period was performed during Weeks 3–18 to equalize the training conditions between subjects. Following the preconditioning period, the participants were submitted at the 8-week intervention with RT plus Cr or Pla supplementation (Weeks 21–28). During this period, dietary intake was monitored over the first and last weeks of intervention (Weeks 21 and 28). Participants were allocated through random number generation. The first measurements took place 1 week before the start of training. The second and third evaluations started on the following Monday and Tuesday after the end of each training phase (Friday), allowing 48–72 hr between the last training and body composition assessment. Figure 1 presents the experimental design.

Participants

Thirty men were recruited through social media and personal invitations to participate in this study. Of these, three dropped out before finishing the intervention due to personal reasons or insufficient training frequency. Thus, a total of 27 men from a university-based population participated in our study and performed both the preconditioning and the supplementation phase. Participants were included in the study if they had no reported disease symptoms, no orthopedic injuries, were not vegetarian/vegan, were not using any nutritional supplements (i.e., protein and Cr powders), were not using anti-inflammatory medicine, and declared that they were free from the use of anabolic steroids. Written informed consent was obtained from all participants after receiving a detailed description of the study procedures. This investigation was conducted according to the Declaration of Helsinki and was approved by the Londrina State University Ethics Committee (protocol number: 028/2012).

Anthropometry

Body mass was measured to the nearest 0.1 kg using a calibrated electronic scale (Balmak, Laboratory Equipment Labstore, Curitiba, Paraná, Brazil) with the participants wearing light workout clothing and no shoes. Height was measured with a stadiometer attached to the scale to the nearest 0.1 cm with subjects standing shoeless. Body mass index was calculated as body mass (in kilograms) divided by the square of height (in meters).

Body Composition

Body composition assessment was conducted in the morning hours. Participants were instructed to urinate ~30 min before the evaluation,

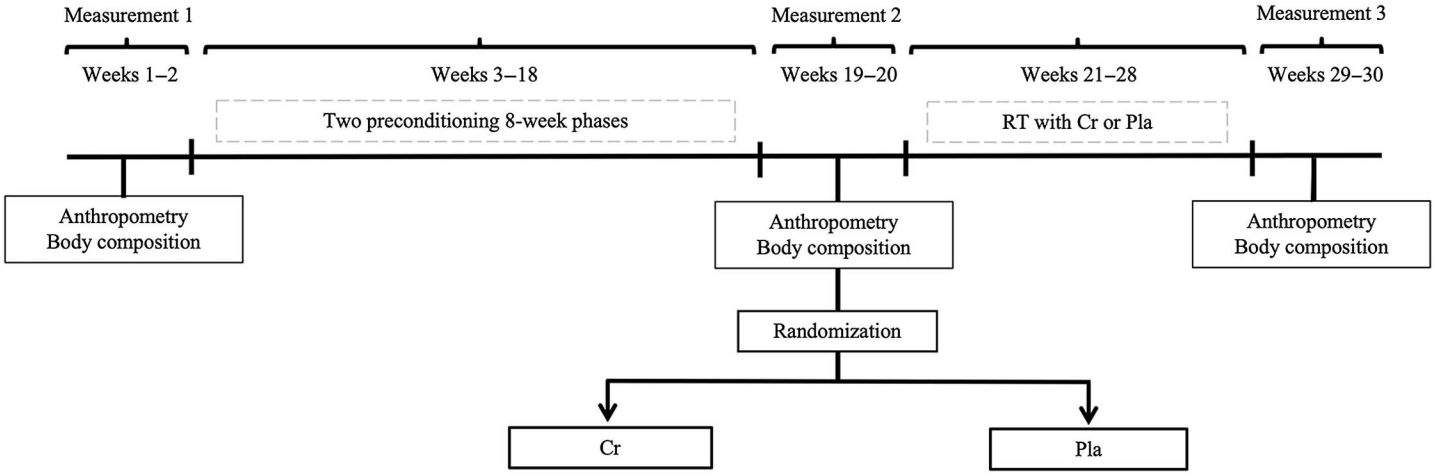

Figure 1 — Experimental design. Cr = creatine; Pla = placebo; RT = resistance training.

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refrain from ingesting food or drink in the previous 4 hr, avoid strenuous physical exercise for at least 24 hr, refrain from consumption of alcoholic and caffeinated beverages for at least 48 hr, and avoid the use of diuretics for at least 7 days prior each assessment.

A spectral bioelectrical impedance device (Xitron Hydra, model 4200; Xitron Technologies Inc., San Diego, CA) was used to estimate TBW, ICW, and ECW. Participants removed all metal objects such as earrings, watches, chains, bracelets from the body before the measurements began. Measurements were performed on a table that was isolated from electrical conductors with subjects lying supine along the table's longitudinal centerline axis, legs abducted at an angle of 45° relative to the body midline, and hands pronated. After cleaning the skin with alcohol, two electrodes were placed on the surface of the right hand and two on the right foot by procedures described by Sardinha et al. (1998). The spectral bioelectrical impedance device was calibrated each day according to the manufacturer's recommendations. The same professional performed the exams in the pre- and postintervention periods, and the intraclass correlation coefficient (ICC) and standard error of measurement (SEM) were: SEM of 0.19 l and ICC = .99 for ICW, SEM of 0.32 l and ICC = .98 for ECW, and SEM of 0.38 l and ICC = .98 for TBW.

Whole-body dual-energy X-ray absorptiometry (Lunar Prodigy, model NRL41990; GE Lunar, Madison, WI) was used to assess appendicular lean soft tissue and body fat. The SMM was estimated by the predictive equation proposed by Kim et al. (2004) as follows: SMM (in kilograms) = (appendicular lean soft tissue × 1.19) – 1.65. Participants removed all metal objects such as earrings, watches, chains, bracelets from the body before the exams began. Scans were performed with the subjects lying in the supine position along the table's longitudinal centerline axis. Feet were taped together at the toes to immobilize the legs while the hands were maintained in a pronated position within the scanning region. Subjects remained motionless during the entire scanning procedure. Both calibration and analysis were carried out by a skilled laboratory technician. Equipment calibration followed the manufacturer's recommendations. The software generated standard lines that set apart the limbs from the trunk and head. These lines were adjusted by the same technician using specific anatomical points determined by the manufacturer. Analyses during the intervention were performed by the same technician, who was blinded to group identity throughout the investigation. The ICC and SEM for SMM were .98 and 0.29 kg, respectively.

Supplementation Protocol

During the supplementation period, the participants were randomly divided into two groups (Pla or Cr supplementation). The supplementation with Cr included a loading period (5 days of 20 g), consumed in four daily doses, followed by a maintenance period (51 days of 3 g), consumed in one dose at breakfast. Maltodextrin was used as a Pla with the same doses and procedures employed in the Cr group. The capsules of Cr or Pla were given to the participants with the exact number of units for the 8 weeks of supplementation. The intake of the supplementation was encouraged by the researchers throughout the study period. Every training day, the participants were asked about their adherence to supplementation and whether they felt any side effects related to its consumption.

Dietary Intake

Participants were instructed by a nutritionist to complete a food record on three nonconsecutive days (two weekdays and one weekend day). Subjects were given specific instructions regarding the recording of portion sizes and quantities to identify all food and fluid intake. Total dietary energy, protein, carbohydrate, and fat content were calculated using Avanutri Processor Nutrition software, version 3.1.4 (Avanutri Equipamentos de Avaliação Ltda, Três Rios, RJ, Brazil). All subjects were asked to maintain their regular diet throughout the study period.

Progressive RT

The progressive RT program was divided into three phases of 8 weeks in a manner designed to induce muscular hypertrophy. All participants were personally supervised by physical education professionals throughout each training session to reduce deviations from the study protocol and to ensure participant safety. During the first two phases, all participants performed a single progressive RT program three times per week on nonconsecutive days (Monday, Wednesday, and Friday), consisting of exercises for the upper limbs, trunk, and lower limbs. Three sets of 8–12 repetition maximum were performed for each exercise. Subjects were instructed to perform each repetition with a 1-s concentric phase followed by a 2-s eccentric phase. The rest period between sets lasted 60–90 s, with a 2–3 min interval between each exercise. Subjects were encouraged to exert maximal effort on all sets. When 12 repetitions were completed for all sets, the training load was increased by 2–5% for upper body exercises and 5–10% for lower body exercises for the next training session.

The first phase of the progressive RT program consisted of nine exercises performed in the following order: (1) bench press, (2) leg press, (3) wide-grip behind-the-neck pulldown, (4) leg extension, (5) side lateral raise, (6) lying leg curl, (7) triceps pushdown, (8) calf press on the leg press machine, and (9) arm curl. In the second phase, the training program was altered, and 11 exercises were performed in the following order: (1) bench press, (2) incline dumbbell fly, (3) wide-grip behind-the-neck pulldown, (4) seated cable rows, (5) seated barbell military press, (6) arm curl, (7) lying triceps press, (8) leg extension, (9) leg press, (10) lying leg curl, and (11) seated calf raise. After the resistance exercises, the abdominal crunch exercise was performed on the floor using the subject's body mass (three sets of 30–50 repetitions in both phases).

In the third phase, wherein participants received supplementation, the RT sessions were performed four times per week, divided into two routines (A and B) wherein Program A was executed on Mondays and Thursdays and was composed of exercises for the chest, shoulders, triceps, and abdominal muscles in the following order: (1) bench press, (2) inclined dumbbell fly, (3) cable cross over, (4) barbell military press, (5) lateral raise, (6) upright row, (7) lying triceps French press, (8) triceps pushdown, and (9) crunch. Program B was conducted on Tuesdays and Fridays, incorporating exercises for the back, biceps, forearm, thigh, and calves in the following order: (1) wide-grip lat pulldown, (2) seated cable row, (3) arm curl, (4) alternating dumbbell curl, (5) wrist curl, (6) squat on a Smith machine, (8) knee extension, (9) leg curl, and (10) seated calf raise.

During this third phase, the subjects performed four sets for all exercises with the load increasing and the number of repetitions simultaneously decreasing for each set (pyramid system). Thus, the number of repetitions used in each set was 12/10/8/6 repetition maximum, respectively, except for the calves (15–20 repetition maximum) and abdominal muscles (150–300 repetitions per session). The load was increased for each set by 2–4 kg for upper

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body exercises and 3–6 kg for lower body exercises. The load progression was planned so that when the participant was able to perform two more repetitions in the last set for a given exercise on two consecutive sessions, the load for the next session was increased 2–5% for the exercises of the upper limbs and 5–10% for the exercises of the lower limbs. Subjects were instructed to perform each repetition with a concentric to eccentric phase ratio of 1:2. The rest period between sets lasted 1–2 min, with a 2–3 min rest interval between each exercise. Adherence to the program was satisfactory with all subjects completing >85% of the total sessions.

Statistical Analysis

Two-way analyses of variance for repeated measures were performed for all comparisons. When the *F* ratio was significant, Fisher's post hoc test was employed to identify where mean differences existed. An independent student's *t* test was performed to determine whether baseline values were significantly different between groups and to compare the percentage changes. The Spearman correlation coefficient was used to determine the relationship between percentage changes in SMM and percentage changes in ICW and ECW. The effect size (ES) was calculated as posttraining mean minus pretraining mean divided by the pooled pretraining *SD* (Cohen, 1992). Relative ES was calculated as the ES difference between conditions (Cr ES – Pla ES). An ES of 0.20–0.49 was considered as small, 0.50–0.79 as moderate, and ≥ 0.80 as large (Cohen, 1992). For all statistical analyses, significance was accepted at $p < .05$. The data were analyzed using SPSS software (version 20.0; SPSS Inc., Chicago, IL).

Results

Table 1 displays the general characteristics of the participants immediately before the presupplementation phase. No differences were observed between groups ($p > .05$). The macronutrients and energy intake at different time points of the study are presented in Table 2. There was no significant ($p > .05$) effect for a group by time interaction and for the main effect of the group for any of the nutritional outcomes analyzed, indicating no between-group differences in the first and last week of training. However, a significant main effect of time ($p < .05$) was observed for protein, carbohydrate, and energy intake, meaning that both groups increased the consumption of these macronutrients throughout the study period.

The body composition values at pre- and postsupplementation phase are presented in Table 3. An interaction was found for body mass, SMM, TBW, and ICW ($p < .05$), wherein increases were greater in the Cr group compared with Pla in body mass (Cr = +3.0%, ES = 0.27; Pla = +0.7%, ES = 0.06), SMM (Cr = +7.1%, ES = 0.71; Pla = +2.8%, ES = 0.28), TBW (Cr = +7.0%, ES = 0.68; Pla = +1.7%, ES = 0.16), and ICW (Cr = +9.2%, ES = 0.81; Pla = +1.6%, ES = 0.14). An effect of time was observed for ECW and percentage of body hydration ($p < .05$), with both groups reaching a similar increase for ECW (Cr = +1.2%, ES = 0.13; Pla = +0.6%, ES = 0.06) and percentage body hydration (Cr = +3.6%, ES = 0.58; Pla = +1.1%, ES = 0.18). The ratio between ICW and SMM did not reach statistical significance ($p > .05$) for any main effects; however, the ratio between ECW and SMM decreased only in the Cr group. No significant effect was observed for body fat ($p > .05$).

Figure 2 displays percentage changes from the pre- to postsupplementation phase in SMM, ICW, and ECW according to group. Increases were higher for Cr than Pla for SMM and ICW ($p < .05$).

A correlation between percentage changes in SMM and ICW was observed ($r = .71$, $p < .001$), indicating that the higher the change in SMM, the greater the change in ICW. However, no significant correlation was observed between percentage changes in SMM and ECW ($r = .12$, $p > .05$). Figure 3 displays the correlation between percentage change from pre- to posttraining in SMM and ICW.

Discussion

The present investigation was designed to investigate the effects of Cr supplementation on changes in SMM, TBW, and its ICW and ECW components when combined with an 8-week RT program. We also endeavored to determine whether this ergogenic aid could influence the ICW/SMM and ECW/SMM ratios in resistance-trained men. The main findings of the present study were: (a) the increase in SMM, ICW, and TBW was higher in the Cr group when compared with Pla; (b) the ICW/SMM ratio did not change due to Cr supplementation combined with RT in resistance-trained men; (c) there was a 50% shared variance between ICW and SMM; and (d) groups did not differ in ECW and percentage body water responses.

Consistent with the current literature, our findings highlight the effectiveness of Cr supplementation in enhancing RT-induced

Table 1 General Characteristics of the Sample

Variables	Creatine (*n* = 14)	Placebo (*n* = 13)	*p*
Age (years)	21.8 ± 4.1	21.7 ± 4.2	.99
Body mass (kg)	70.1 ± 7.4	68.0 ± 8.4	.48
Height (cm)	173.0 ± 4.5	175.5 ± 7.3	.30
Body mass index (kg/m^2)	23.3 ± 2.1	22.0 ± 2.3	.13

Note. Data are presented as mean ± *SD*.

Table 2 Energy and Macronutrients Intake at Different Moments According to Group

	Creatine (*n* = 14)		Placebo (*n* = 13)		Effects		
Variables	Pre	Post	Pre	Post	Group	Time	Interaction
Energy (kcal·kg^{-1}·day^{-1})	28.65 ± 7.3	31.25 ± 10.0*	27.36 ± 8.2	35.44 ± 10.1*	.65	.03	.27
Protein (g·kg^{-1}·day^{-1})	1.28 ± 0.4	1.51 ± 0.6*	1.22 ± 0.4	1.67 ± 0.6*	.79	.03	.46
Carbohydrate (g·kg^{-1}·day^{-1})	3.94 ± 1.4	4.02 ± 1.5*	3.88 ± 1.2	4.92 ± 1.37*	.48	.04	.89
Lipid (g·kg^{-1}·day^{-1})	0.93 ± 0.2	1.01 ± 0.3	0.93 ± 0.3	1.00 ± 0.3	.99	.38	.93

Note. Data are presented as mean ± *SD*.
*$p < .05$ versus pre.

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Table 3 Participants' Scores at Pre and Post the 8-Week Intervention Period

Variables	Creatine ($n = 14$)		Placebo ($n = 13$)		Relative effect size	Relative differences	Interaction p value
	Pre	Post	Pre	Post			
Body mass (kg)	70.1 ± 7.4	72.2 ± 6.9	67.9 ± 8.4	68.4 ± 8.1	0.20	2.3	<.01
Appendicular lean soft tissue (kg)	28.7 ± 2.5	30.6 ± 2.6*	28.3 ± 4.0	29.1 ± 4.0*	0.43	4.3	<.001
SMM (kg)	32.5 ± 2.5	34.8 ± 2.6*,**	32.1 ± 4.0	33.0 ± 4.0*	0.43	4.3	<.001
Body fat (%)	13.8 ± 5.7	14.0 ± 5.8	12.5 ± 6.5	12.8 ± 6.1	–0.02	–1.0	.93
Total body water (L)	43.0 ± 3.1	46.0 ± 3.3*,**	41.8 ± 5.7	42.5 ± 4.9*	0.52	5.3	<.01
ICW (L)	26.2 ± 2.1	28.6 ± 2.4*,**	25.3 ± 3.8	25.7 ± 3.2*	0.68	7.6	<.01
ECW (L)	17.1 ± 1.1	17.3 ± 1.2*	16.7 ± 2.1	16.8 ± 1.9*	0.06	0.6	.87
ICW/SMM (L/kg)	0.81 ± 0.06	0.82 ± 0.05	0.79 ± 0.07	0.78 ± 0.61	0.31	2.5	.14
ECW/SMM (L/kg)	0.54 ± 0.02	0.52 ± 0.02*	0.53 ± 0.01	0.53 ± 0.02	–1.00	–4.8	<.001
Percentage of body water	61.7 ± 3.2	63.9 ± 4.1*	61.5 ± 4.4	62.2 ± 4.1*	0.39	2.4	.10

Note. Data are expressed as mean ± *SD*. Effect size classification = 0.20–0.49 small, 0.50–0.79 moderate, and ≥0.80 large. Relative effect size = effect size of creatine group minus effect size of Pla group; relative differences = Δ% of creatine group minus Δ% of Pla group; ICW = intracellular water; ECW = extracellular water; SMM = skeletal muscle mass.
*$p < .05$ versus pre. **$p < .05$ versus Pla.

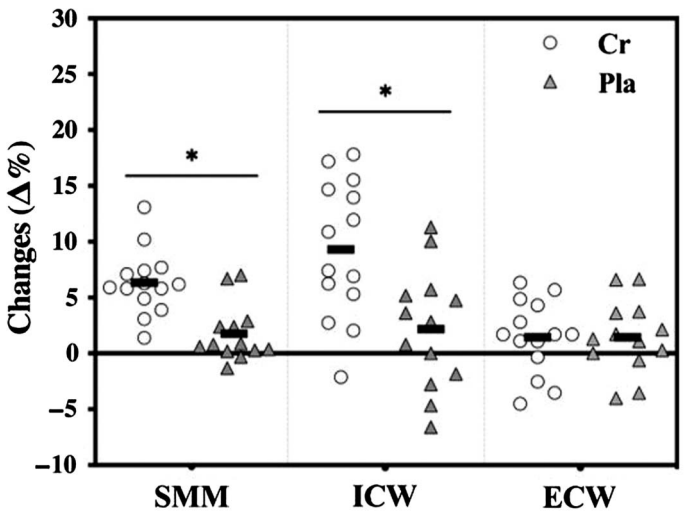

Figure 2 — Percentage of changes from pre- to posttraining (8 weeks) on SMM, ICW, and ECW in resistance-trained men according to groups (Pla, $n = 13$; Cr, $n = 14$). Data are expressed as mean and *SD*. SMM = skeletal muscle mass; Cr = creatine; Pla = placebo; ICW = intracellular water; ECW = extracellular water. *$p < .05$ versus Pla.

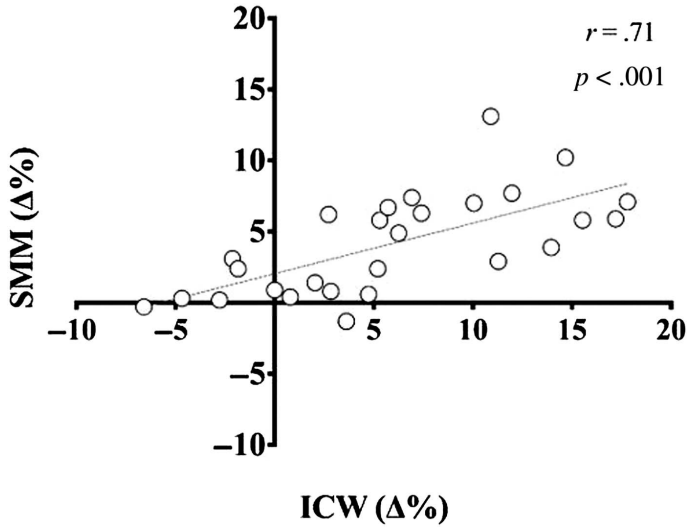

Figure 3 — Association between percentage changes from pre- to posttraining (8 weeks) on SMM and ICW in resistance-trained men ($n = 27$). SMM = skeletal muscle mass; ICW = intracellular water.

gains in muscle mass (Kerksick et al., 2018). Among the proposed mechanisms for the effects of Cr supplementation in promoting muscle mass accretion, an increase in intracellular hydration has been furthered as a likely contributor. Cr is cotransported with Na^+ ions across the sarcolemma, which initiates an influx of Cl^- and water to balance electroneutrality and osmolality (Odoom et al., 1996), thus contributing to the movement of ECW to the ICW compartment. Moreover, Cr has been shown to enhance glycogen storage (van Loon et al., 2004). This may further heighten intracellular hydration, given that each gram of glycogen attracts approximately 3–4 g of water (Olsson & Saltin, 1970). The observed increase in SMM in both groups, particularly in the group supplemented with Cr, was associated with ICW expansion. This association is expected as ICW may be regarded as a functional water compartment. There is substantial evidence to suggest that cellular hydration status is an essential factor controlling cellular protein turnover (Häussinger et al., 1993). In fact, cell swelling acts as an anabolic signaling stimulus, mediating muscle protein synthesis via the mitogen-activated protein kinase signaling cascade (Niisato et al., 1999). Moreover, Silva et al. (2014) found that athletes who increase ICW over the season improved power and strength, suggesting that ICW may be an indicator of performance.

The more significant increase in ICW from Cr supplementation combined with RT has been previously reported by Francaux and Poortmans (1999) in a cohort of untrained men submitted to 9 weeks of supplementation or Pla. Alternatively, it has also been reported that Cr supplementation combined with RT did not induce a significant increase in ICW in a cohort of resistance-trained men and women (Powers et al., 2003). The reason that our results differ from this latter investigation is not clear. It may be, at least in part, related to the fact that we employed an adaptation phase wherein all participants performed the same program for 16 weeks, whereas

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the previous study initiated the study period immediately upon consent. It is conceivable that novel aspects of the training protocol confounded results in the study by Powers et al. (2003) and, thus, masked potential changes in ICW between conditions; the novelty factor would not apply to our study given the imposed acclimation phase. This hypothesis warrants further investigation.

Despite the greater ICW increase in the Cr group compared with Pla, the ICW/SMM ratio did not change in any group. We did not attempt to measure intramuscular Cr stores, but similar loading/maintenance protocols have shown these storage depots to be significantly elevated after supplementation (Preen et al., 2003). Given that Cr acts as an osmolyte, it can be surmised that, perhaps, similar amounts of contractile hypertrophy occurred between conditions and that the additional increases in SMM observed in the Cr group were attributed mainly to "sarcoplasmic hypertrophy" as previously described in the literature (Haun, Vann, Osburn, et al., 2019). Further research is warranted by employing the biopsy technique to determine subfractional changes within the muscle when Cr supplementation is combined with regimented RT.

Another important finding of our investigation is that Cr supplementation did not promote changes in ECW. Despite the lack of effect, the RT protocol did elicit an increase in this water compartment. This result may be related to the rise of SMM, because the water in the skeletal muscle is distributed to both ICW and ECW (the sum of interstitial fluid and blood plasma; Haun, Vann, Roberts, et al., 2019). Thus, an increased SMM is seemingly accompanied by increased ECW and ICW. Hence, it is reasonable to speculate that the increases in SMM were associated with an expanded extracellular fluid balance.

It is noteworthy that the ECW/SMM ratio decreased only in the Cr group. Recently, it has been reported that the relative expansion of ECW above that of ICW may have an adverse effect on muscle quality (Yamada et al., 2017). Based on this information, our finding of a reduced ECW/SMM ratio following Cr supplementation compared with Pla may confer a beneficial effect on SMM quality. This hypothesis warrants further study.

To the best of our knowledge, this is the first investigation to analyze the response of SMM and ICW to the combination of RT and Cr supplementation in resistance-trained men. Although our study has notable strengths—(a) a randomized, double-blinded, and Pla-controlled trial; (b) a preconditioning-RT protocol to reduce confounding from novelty; and (c) validated testing measures—it is worth mentioning some possible limitations. First, we did not prescribe a dietary protocol for participants throughout the experiment but, rather, only monitored dietary consumption during the first and last weeks of training. Therefore, we were unable to verify whether the diets they followed may have unduly affected our findings. However, all participants were asked continuously not to change their eating habits during our experiment, and random group assignment should, conceivably, diminish potential confounding in this regard. Moreover, pre- and poststudy nutritional assessment did not reveal significant between-group differences in energy or macronutrient intake. Second, the duration of the supplementation phase of our investigation was relatively short (8 weeks). Therefore, it is not clear whether these responses would occur similarly after a longer period of RT and Cr supplementation. We should also note that dual-energy X-ray absorptiometry scans were performed at least 4 hr after fluid and food ingestion. An overnight fast would be the most effective way to minimize measurement errors in lean soft tissue determination (Nana et al., 2015). Also, biopsies would have been beneficial to help assure that Cr supplementation was successfully administrated. Finally, our study is specific to resistance-trained young men and should not necessarily be generalized to other populations, including women, youth, and older individuals.

Conclusion

This investigation suggests that Cr supplementation combined with RT induces higher increases in SMM and ICW but does not change the ratio between SMM and ICW in resistance-trained young men. These inferences' strength of confidence must be balanced with the limitations mentioned above of the study design. From a practical standpoint, our results reinforce that Cr supplementation combined with RT is an effective ergogenic aid that promotes increased muscle mass in resistance-trained men. According to our findings, changes in SMM and ICW occur similarly, with no differences in the ratio between these two components, even with Cr supplementation. Therefore, resistance-trained men who use Cr will be able to maximize muscle mass gains, accompanied by increased cellular hydration, when combined with RT.

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